

DEVA ANAND

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Education

MS, Computer Science, University of Massachusetts Amherst 2025–2027

Advanced NLP, Advanced Machine Learning, Optimization in CS, Systems for Data Science, Performance Engineering

BE, Computer Science and Engineering, Vels Institute of Science, Technology and Advanced Studies 2019–2023

Professional Experience

Wysa, Backend Engineer | Bengaluru, India Jul 2023 – Jul 2025

- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clinical clients**, processing **10,000+ monthly triage submissions** through REST integrations with Mayden's iaptus enterprise platform; contributed to **\$340K+** in revenue.
- Architected a full-stack ML training-data annotation platform (React, Node.js, MongoDB) used daily by the AI team, replacing spreadsheet-based labeling with structured role-based data workflows that cut manual annotation by **100+ hours/month** and lifted labeling throughput by **60%**.
- Implemented backend security controls, **AWS KMS** encryption for PII/clinical data, 90-day retention, auth hardening, and rate limiting, and remediated **20+ VAPT findings**.
- Served as primary technical POC for the eTriage backend, ran knowledge transfer onboarding a second developer to on-call readiness in **~6 weeks**, and piloted a COO-sponsored GitHub Copilot experiment for AI-assisted development.

Cario Growth Services, Backend Developer Intern | Chennai, India Nov 2022 – May 2023

- Designed and shipped **20+ RESTful APIs** (Node.js, Fastify, PostgreSQL) with validation, clear contracts, and backward-compatible changes used in production by the frontend team.
- Migrated core repositories from JavaScript to TypeScript and shipped a production LLM auto-commenting feature (**4-bit quantized Llama-7B via llama.cpp**) with per-persona prompts, async queued generation, and content-safety filtering.

Freelance / Open Source, Full-Stack Engineer | Remote Jul 2022 – Aug 2023

- Built and deployed **Underdogs Fitness** (Website), a production MERN gym-management platform with Stripe + manual-cash hybrid payments, cron automation, and multi-branch support; live **3+ years**.
- Delivered a second MERN client engagement (US-based inventory management system) end-to-end as a paid freelance build.

Projects

AgenticSearch – Provenance-First Entity Discovery (GitHub | Demo) Python, FastAPI, OpenAI / Groq, SQLite

- Built a multi-stage agentic web-discovery service turning open-ended queries into structured entity tables via typed retrieval planning, reranking, and deterministic extraction with LLM fallback and **cell-level provenance**; hardened with **150+ tests** and provider routing across LLM backends.

KG2RAG-Enhanced – Multi-Hop QA on HotpotQA (GitHub) Python, Ollama / LLaMA-3, sentence-transformers, RRF

- Built the multi-view retrieval component of a KG-guided RAG pipeline (query-side decomposition, RRF fusion, cross-encoder reranking, MMR), lifting supporting-fact recall **+2.68%** over the KG²RAG baseline on a 4,905-question evaluation.

cf-ai-research-scout – Edge-Native AI Assistant (GitHub) TypeScript, Cloudflare Workers AI, Durable Objects, D1

- Built an edge-native AI research assistant where an agent fetches web sources and streams multi-step research digests over WebSocket, with durable workflow orchestration and D1-backed state surviving process restarts and tool failures.

Publications

- "Alzheimer's Disease Classification using Transfer Learning," IEEE CONIT 2023, first author, deep transfer learning on neuroimaging data. (Paper)

Skills

Languages & Databases: Python, TypeScript, JavaScript, SQL, C++, Java, Bash, PostgreSQL, MongoDB, SQLite

Frameworks / Tools: FastAPI, Node.js, Express, React, REST APIs, Docker, Git, GitHub Actions, Cloudflare Workers

ML & AI: LLM agents, RAG, prompt engineering, PyTorch, Hugging Face Transformers, sentence-transformers, evaluation

Retrieval / Search: RRF, cross-encoder reranking, MMR, knapsack token budgeting, Brave Search, cell-level provenance

Cloud / Platforms: Azure, AWS, GCP, Linux, MongoDB Atlas

Achievements

- Selected for **GirlScript Summer of Code 2023**; contributed to Linkfree, Freehit, and ProjectsHut. - Tutored **30+ underprivileged students** in programming at Sayur, a non-profit in Tamil Nadu.